Type MS · RAM, ROM, virtual memory and flash memory

Answer 1 of 7 · Question 2 part b.i · 2 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part b.i in this mark scheme extract

Number			
2(a)(i)	Award **one** mark for any of the following: • To perform end **user** related tasks (1) • To help the user be more productive/creative/communicative (1)	Do not award trade names or generic categories of software Allow software examples such as word processor if it is clear that it is an end-user task. Do not award responses related to tasks that could be performed by pen/paper.	1
2(a)(ii)	Award **one** mark for any of the following up to a maximum of **three** marks: • Process management (1) • Memory management/virtual memory (1) • File management (1) • Peripheral/hardware management (1) • Network management (1) • Security management (1) • User management (1)	Allow equivalence to management such as 'scheduling processes', 'Virtual memory organisation', but do not award terms in isolation. Do not award marks for types of OS such as multitasking Do not award UI as given in question.	3

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · RAM, ROM, virtual memory and flash memory

Answer 1 of 7 · Question 2 part b.i · 2 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part b.i in this mark scheme extract

2(b)(i)	Award up to **two** marks for a linked explanation such as: • RAM is volatile/its contents will be lost when the power is switched off (1) ROM (is permanent) is required to store the BIOS/firmware/to boot up the computer (1) • The contents of ROM cannot be changed (1) RAM is required for actively executing programs and data (1)	Both RAM and ROM must be addressed for 2 marks.	2
2(b)(ii)	Award **one** mark for any of the following up to a maximum of **three** marks: • A platter is used (1) • (The platter) is divided into tracks / sectors (1) • A read/write arm/head is moved (to the track required) (1) • A read/write head reads the values (from the surface) (1) • One polarity represents 1 and other polarity represents 0 (1)	Accept North/South for polarity and representation of binary state.	3

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · RAM, ROM, virtual memory and flash memory

Answer 1 of 7 · Question 2 part b.i · 2 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part b.i in this mark scheme extract

<table>
<tr>
<td>2(c)</td>
<td>Award one mark for each correct row:
<table>
<tr><th>Stage</th><th>Sequence Number</th></tr>
<tr><td>The instruction is placed onto the data bus</td><td>2</td></tr>
<tr><td>The Arithmetic and Logic Unit (ALU) performs a calculation</td><td>4</td></tr>
<tr><td>The address of the next instruction is placed onto the address bus</td><td>1</td></tr>
<tr><td>The instruction is decoded by the control unit</td><td>3</td></tr>
</table>
</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>2(d)</td>
<td>The only correct answer is A

B is not correct because the control bus is bidirectional
C is not correct because the system bus is a collective term that contains both unidirectional and bidirectional buses
D is not correct because the data bus is bidirectional</td>
<td></td>
<td>1</td>
</tr>
<tr>
<td colspan="2"></td>
<td>Total for question 2</td>
<td>14</td>
</tr>
</table>

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · RAM, ROM, virtual memory and flash memory

Answer 1 of 7 · Question 2 part b.i · 2 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · RAM, ROM, virtual memory and flash memory

Answer 2 of 7 · Question 1 part d · 2 marks

2019 JUN · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

1(a)	Award **one** from: Input, process, output in this order – **3 marks** Input, process, output in any order – **2 marks** At least one of input, process, output – **1 mark**		3

Question Number	Answer	Additional Guidance	Mark
1(b)	B: Central processing unit **The only correct answer is B** **A** is not correct as this is memory only **C** is not correct as this is memory only **D** is not correct as this is used only for transporting memory addresses		1

Question Number	Answer	Additional Guidance	Mark
1(c)	• Data (1) • Instructions (1)		2

Question Number	Answer	Additional Guidance	Mark
1(d)	Any **two** from: 1. Virtual memory (VM) is used when RAM becomes full (1) (to hold all programs and data). 2. Virtual memory is used as (an extension to) main memory/RAM / works like RAM. (1) 3. Virtual memory is stored/created on (internal) secondary storage/HDD/SSD. (1) 4. Virtual memory is used as temporary storage. (1) 5. Instructions and data not currently being used are transferred from RAM to VM/HDD. (1) 6. When needed again, instructions and data are transferred back to RAM. (1)		2

Type MS · RAM, ROM, virtual memory and flash memory

Answer 2 of 7 · Question 1 part d · 2 marks

2019 JUN · Paper 01 · extract 2 of 3 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(e)	Any **two** from: 1. The disc surface is made out of polycarbonate. (1) 2. A laser (beam) is used. (1) 3. Data is stored as a series of tiny grooves arranged in tracks on the surface of the disc. (1) 4. Bumps/lands/flats/troughs/pits/dips imprinted/burnt on the disc. (1) 5. Bumps and flats (etc.) represent 1 and 0 (bits) / the reflection (of the laser) represents 1 and 0 (bits). (1)		2

Question Number	Answer	Additional Guidance	Mark
1(f)	D: System software **The only correct answer is D** **A** is not correct because this category is purely for software that is used for security purposes **B** is not correct because this category is designed for end user tasks rather than for managing the computer itself **C** is not correct because this software is used to translate a high level program into machine code		1

Type MS · RAM, ROM, virtual memory and flash memory

Answer 2 of 7 · Question 1 part d · 2 marks

2019 JUN · Paper 01 · extract 3 of 3 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · RAM, ROM, virtual memory and flash memory

Answer 3 of 7 · Question 3 part b.ii · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number		Guidance	
3(a)	Award **two** marks for an explanation that includes hardware **and** software. • The laptop is/contains the hardware (1) and the software allows the user to interact with it (1) • Hardware gives input and output devices (1) /memory, storage, cpu, processing ability • Software allows instructions to be passed to the hardware (1) so that it can carry out tasks (1) • Software is the set of instructions (1) that controls the hardware (1)		2

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	Award **two** marks from: • Lighter/thinner (1) • Less heat generated (1) • Faster access speed / faster booting of operating system / faster data transfer/read/write speeds (1) • Silent operation (1) • Less power required / longer battery life (1) • More robust (due to no moving parts) (1)		2
3(b)(ii)	Award **two** marks for a linked explanation such as: • Flash memory is non-volatile (1) whereas RAM is volatile (1) • Flash memory persists even when power is lost (1) whereas RAM clears its data whenever power is lost (1) • Flash memory is used for storage (1) whereas RAM is used in performing operations on data taken from storage (1)		2

Type MS · RAM, ROM, virtual memory and flash memory

Answer 3 of 7 · Question 3 part b.ii · 2 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	Award **two** marks for a linked explanation such as: • Cache memory stores active/commonly used instructions / cache is a temporary storage (1) and is used to speed up processing. (1) stop data transfer bottlenecks between RAM and CPU (1)		2

Question Number	Answer	Additional Guidance	Mark
3(d)	Award **one** mark for two correct rows. Award **two** marks for four correct rows.		2

Device	Input	Output	Both
Headset			✓
Mouse	✓		
Printer		✓	
Touch screen			✓

Question Number	Answer	Additional Guidance	Mark
3(e)	Award **two** marks from: • Antivirus (1) • Anti-spyware (1) • (Software) firewall (1)	Do not award tradenames	2

Question Number	Answer	Additional Guidance	Mark

Type MS · RAM, ROM, virtual memory and flash memory

Answer 4 of 7 · Question 5 part d.iii · 2 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part d.iii in this mark scheme extract

5(a)(i)	Any **two** from: • It will interpret/analyse patient input to identify symptoms (1) and match the symptoms to (possible) illnesses (1) • It will match symptoms to possible illnesses (1) and give the most likely/probable illness (1) • It will match symptoms to possible illnesses (1) and ask further questions to narrow it down (1) • It will match symptoms to possible illnesses (1) by searching/using a database/other data store (1)		2
5(a)(ii)	Award **one** mark from: 1. May not have access to the internet (1) 2. May not have access to a device (1) 3. May not want to use it (1) 4. May not have the technical knowledge to use it (1) 5. May have a physical disability that stops them from using the service (1) 6. May not want to disclose personal information (1) 7. hacking or security or data issue (1) 8. May not trust the AI/system (1) 9. Can't take physical measurements eg. blood pressure (1) 10. may want a real person (1)		1

Type MS · RAM, ROM, virtual memory and flash memory

Answer 4 of 7 · Question 5 part d.iii · 2 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part d.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **one** mark for: • Local area network / LAN /VLAN		1
5(b)(ii)	Award **one** mark for: • Wide area network / WAN		1

Question Number	Answer	Additional Guidance	Mark
5(c)	**The only correct answer is C** ***A*** *is not correct because this is phishing* ***B*** *is not correct because this is shoulder surfing* ***D*** *is not correct because this is pharming*		1

Type MS · RAM, ROM, virtual memory and flash memory

Answer 4 of 7 · Question 5 part d.iii · 2 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part d.iii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award up to two marks for a linked explanation such as:

• Robust/less likely to break it if dropped / quieter than a mechanical hard drive (1) because it doesn't have any moving/mechanical parts (1)
• Smaller/thinner/lighter than a device with HDD (1) so easier to carry/more portable / laptops have less room in them for drives/componants(1)
• SSD uses less power (1) so battery will last longer (1)
• SSD is faster (1) so can access data/records more quickly (1)</td><td></td><td>2</td></tr>
<tr><td>5(d)(ii)</td><td>Any two from:

• Flash memory chips are used (1)
• Chips have (floating gate) transistors/electron pools/charge traps (1)
• Gates/electron pools/charge traps hold an electrical charge (1)
• Charge remains even when no power (1)
• No charge/empty pool/trap represents data/1 (1)
• A charge/full pool/trap represents no data/0 (1)
• Data is stored in blocks and pages (1)</td><td></td><td>2</td></tr>
<tr><td>5(d)(iii)</td><td>Award one mark for each of:

<table><tr><th>Description</th><th>RAM</th><th>ROM</th></tr><tr><td>Stores the boot up sequence</td><td></td><td>✓</td></tr><tr><td>The contents are lost when the laptop is shut down</td><td>✓</td><td></td></tr></table></td><td></td><td>2</td></tr>
<tr><td colspan="3">Total for question 5</td><td>12</td></tr>
</table>

Type MS · RAM, ROM, virtual memory and flash memory

Answer 4 of 7 · Question 5 part d.iii · 2 marks

2022 JUN · Paper 01 · extract 4 of 4 · find part d.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · RAM, ROM, virtual memory and flash memory

Answer 5 of 7 · Question 5 part b.ii · 1 mark

2023 JUN · Paper 01 · extract 1 of 8 · find part b.ii in this mark scheme extract

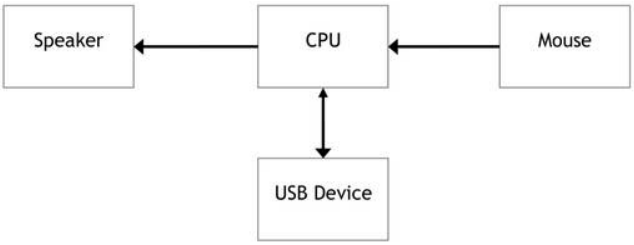
<table>
<tr>
<td>5(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Req.</th><th>Example</th></tr>
<tr><td>Input</td><td>1. Enter the direction the bee will face (1)
2. Enter the number of steps that the bee will move (1)</td></tr>
<tr><td>Process</td><td>Calculate the path the bee will move along to its new position</td></tr>
<tr><td>Output</td><td>3. The bee appears in its new position (1)
4. The bee flies to its new position (1)
5. The bee faces its new direction (1)</td></tr>
</table>
</td>
<td>Allow direct instructions as input e.g. ‘Turn East’ and consequences for output e.g. ‘Bee turns to face East’ if clear.

The input/output do not have to be directly linked. E.g. Input Turn West, Output Has Moved 20 steps</td>
<td>2</td>
</tr>
</table>

Type MS · RAM, ROM, virtual memory and flash memory

Answer 5 of 7 · Question 5 part b.ii · 1 mark

2023 JUN · Paper 01 · extract 2 of 8 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5b(i)	Award one mark for each: • Line between Mouse and CPU with arrow at CPU end only (1) • Line between CPU and Speaker with arrow at Speaker end only (1) • Line between CPU and USB Device with arrows on both ends (1) 	Allow two independent back/forth arrows between CPU and USB in place of a single bi-directional arrow. Ignore any additional lines.	3

Type MS · RAM, ROM, virtual memory and flash memory

Answer 5 of 7 · Question 5 part b.ii · 1 mark

2023 JUN · Paper 01 · extract 3 of 8 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(ii)	**The only correct answer is B** ***A*** *is not correct because a* ***firewall*** *is an application that is loaded after the device is booted* ***C*** *is not correct because* ***the operating system*** *is loaded after the device is booted* ***D*** *is not correct because* ***the user interface*** *is loaded after the device is booted*		1

Question Number	Answer	Additional Guidance	Mark
5(b)(iii)	Award up to **two** marks for a linked explanation, such as: • Cache makes up for the difference in speed of two devices (1) because it is very high-speed memory (1) • Cache is high-speed memory (1) which acts as a buffer between a faster and a slower device (1) • It speeds up processing (1) because active/commonly used instructions and data are stored nearer to the CPU / the CPU doesn't have to wait while instructions/data are fetched from RAM (1)		2

Type MS · RAM, ROM, virtual memory and flash memory

Answer 5 of 7 · Question 5 part b.ii · 1 mark

2023 JUN · Paper 01 · extract 4 of 8 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)	Award up to **two** marks for a linked description, such as: • Uses a sensor (1) to check the water level in the machine (1) • Uses a thermometer (1) to check the water temperature (1) • Uses a timer (1) to control the time required for the wash cycle/rinse/drain/spin (1) • Uses a motor (1) to rotate the drum (1)	The first mark is for identification and the second is for the linked description. It is possible to gain one mark for a description without identifying the sensor/actuator.	2

Type MS · RAM, ROM, virtual memory and flash memory

Answer 5 of 7 · Question 5 part b.ii · 1 mark

2023 JUN · Paper 01 · extract 5 of 8 · find part b.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award one mark for each two correct rows.

<table>
<tr><th>A</th><th>B</th><th>C</th><th>A AND C</th><th>(A AND C) OR B</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table>
</td><td></td><td>3</td></tr>
</table>

Type MS · RAM, ROM, virtual memory and flash memory

Answer 5 of 7 · Question 5 part b.ii · 1 mark

2023 JUN · Paper 01 · extract 6 of 8 · find part b.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(ii)</td><td>Award one mark for each of:

• M OR P (1)
• NOT C (1)
• <expr1> AND <expr2> (1)

Example statements:

(M OR P) AND (NOT C)
(M OR P) AND NOT C
(M AND NOT C) OR (P AND NOT C)
M AND NOT C OR P AND NOT C

For information only:

<table>
<tr><th>M</th><th>P</th><th>C</th><th>M OR P</th><th>NOT C</th><th>(M OR P) AND (NOT C)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table>
</td><td>Third bullet awarded even if first two are not

Order of precedence is NOT, AND, OR

Check brackets carefully, if provided</td><td>3</td></tr>
</table>

Type MS · RAM, ROM, virtual memory and flash memory

Answer 5 of 7 · Question 5 part b.ii · 1 mark

2023 JUN · Paper 01 · extract 7 of 8 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. The operating system uses a scheduler to control processes (1) 2. The operating system holds processes in a queue (1) 3. Some processes may be given higher priorities than others (1) 4. Each process gains accesses to the CPU for a short time / time slice to execute (1) 5. Processes are swapped to/from (the queue / CPU)	Do not award responses about memory management, as the question is about sharing the CPU only	2

Type MS · RAM, ROM, virtual memory and flash memory

Answer 5 of 7 · Question 5 part b.ii · 1 mark

2023 JUN · Paper 01 · extract 8 of 8 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · RAM, ROM, virtual memory and flash memory

Answer 6 of 7 · Question 2 part b.i · 1 mark

2025 JUN · Paper 01 · extract 1 of 3 · find part b.i in this mark scheme extract

2(a)	Award **one** mark for any of the following up to a maximum of **two** marks: • (Touch)screen (1) • Speaker (1) • Vibration (motor) (1) • Flashlight/LED light/torch (1) • NFC chip (1)	**Accept:** • display for screen **Do not accept:** • monitor • printer • headset • other peripheral devices	**2**
Question	Answer	Additional Guidance	Mark
2(b)(i)	**The only correct answer is D** *A is not correct because cache is located closer to the CPU* *B is not correct because random-access memory is a physical card on the motherboard* *C is not correct because read-only memory cannot be written to*		**1**
Question	Answer	Additional Guidance	Mark
2(b)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • Often held/stored remotely/stored on the internet (1) • Held/managed by a third party (1) • Requires internet connectivity to access (1)		**2**

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Type MS · RAM, ROM, virtual memory and flash memory

Answer 6 of 7 · Question 2 part b.i · 1 mark

2025 JUN · Paper 01 · extract 2 of 3 · find part b.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark
2(c)(i)	Award **one** mark for any of the following up to a maximum of **three** marks: • Clock speed/number of cycles per second (1) • Number of processor cores (1) • Size of cache (memory) (1) • Type of cache (1) • Width of data bus/word length (1) • Width of address bus (1) • Whether the CPU is overclocked or not (1) • Pipelining architecture (1) • Thermal throttling / cooling system (reducing clock speed) (1)	**Accept:** • units of clock speed e.g. GHz for MP1 • width of bus for either MP5 or MP6 but not both **Do not accept:** cache without clarification of size or type	3
Question	Answer	Additional Guidance	Mark
2(c)(ii)	Award up to **two** marks for a linked description such as: • The CPU places an address on to the address bus (1) to specify the location in main memory (1) • (The address bus) contains the location in RAM (1) to read from/write to (1) • The value in the memory address register (MAR) is placed on the address bus (1) to access the correct location in RAM (1) • The address bus carries the memory location (1) of the current/next instruction (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
Question	Answer	Additional Guidance	Mark
2(d)	Award **one** mark for any of the following up to a maximum of **two** marks: • Sub-atomic particles (1) • Qubit(s) (1) • Entanglement (1) behaviour of a pair of qubits can influence each other • Quantum parallelism (1) - allowing a quantum computer to evaluate multiple computational paths simultaneously	**Do not accept:** 'Qubits can exist in a state of 1 and 0 at the same time' as this is the same as 'superposition' given in the question	2
		Total for question 2	**12**

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Type MS · RAM, ROM, virtual memory and flash memory

Answer 6 of 7 · Question 2 part b.i · 1 mark

2025 JUN · Paper 01 · extract 3 of 3 · find part b.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · RAM, ROM, virtual memory and flash memory

Answer 7 of 7 · Question 4 part c.ii · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part c.ii in this mark scheme extract

4(a)(i)	Award **one** mark for any of the following up to a maximum of **two** marks: • It has high bandwidth/transmission speed (1) • It has low latency (1) • It has low energy consumption (1) • It has high capacity (1) • It has high security (1)	Allow 'more users can be connected' for MP4	**2**
Question	Answer	Additional Guidance	Mark
4(a)(ii)	Award up to **two** marks for a linked description such as: • Biometric identifiers (fingerprint, iris, face) (1) matched against those stored in the app/database (1) • Two-factor authentication / 2FA (1) requiring users to enter a code delivered to a separate system (1) • Users can be asked for memorable information/security question (1) that only they will know the answer to (1)	System in MP2b may be on a separate device or a separate app	**2**
Question	Answer	Additional Guidance	Mark
4(a)(iii)	Award **one** mark for any of the following up to a maximum of **two** marks: • To identify any bad programming practices/shortcuts/bugs (1) • To highlight code vulnerabilities / enhance security (1) • To make sure code is robust/will deal with exceptions (1) • To make sure that good programming practices/company policies are maintained (1) • To assess the efficiency of the code (1) • To improve code quality (1) • To share good practice/new techniques (1)		**2**

Type MS · RAM, ROM, virtual memory and flash memory

Answer 7 of 7 · Question 4 part c.ii · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part c.ii in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(b)</td><td>Award one mark for each correct row:
<table>
<tr><th></th><th>8-bit binary pattern</th><th>Denary value</th></tr>
<tr><td>Lowest value</td><td>1111 1111</td><td>-127</td></tr>
<tr><td>Highest value</td><td>0111 1111</td><td>(+)127</td></tr>
</table></td><td></td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(i)</td><td>Award one mark for each correct column:
<table>
<tr><th>O</th><th>S</th><th>P</th><th>O AND S</th><th>NOT P</th><th>(O AND S) AND (NOT P)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
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</table></td><td></td><td>3</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>4(c)(ii)</td><td>Award up to two marks for a linked explanation such as:
• ROM is non-volatile/permanent (1) therefore it retains its contents when the power is switched off / data won't be lost when power is lost (1)
• The contents of ROM cannot be changed (1) which makes it suitable for storing the manufacturer's instructions (1)</td><td>Explanation can be a combination of different marking points. E.g. MP1a and MP2b
Do not accept MP1a and MP2a or MP1b and MP2b</td><td>2</td></tr>
<tr><td colspan="3">Total for question 4</td><td>13</td></tr>
</table>

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Type MS · RAM, ROM, virtual memory and flash memory

Answer 7 of 7 · Question 4 part c.ii · 2 marks

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Question	Answer	Additional Guidance	Mark